

Technology and innovation is changing the face of Australia's forestry industries.

Watch

[Innovation and technology \(transcript\) DOCX](#)  [25 KB]

New ways to manage forests

Innovative technologies are helping the industry to better manage Australia's forests and produce more from wood.

These technologies include:

- drones
- lasers
- scanners
- infra-red sensors
- 3D sensing and imaging
- advances in tree-breeding.

New technologies are helping to manage Australia's forests more sustainably. For instance, drones are used to obtain accurate forest data to improve forest planning and management processes.

New ways to track illegal logging

DNA also has its place in forest management. By using DNA fingerprinting of wood, we can verify the legality of wood products. Individual logs and wood products can be traced back to the forest of origin.

This technology is helping Australia combat illegal logging and associated trade. It benefits the environment and businesses that are doing the right thing in Australia and around the world.

New materials

Technological advances are also unlocking an array of new and exciting materials that can be derived from trees.

These include:

- engineered wood products
- pelletised wood
- chemicals from wood
- nano-cellulose.

Forestry plays an important role in mitigating climate change. Australia's forests store nearly 22 billion tonnes of carbon, in addition to carbon being stored in wood products.

By developing biofuels derived from wood, the forestry industry is getting closer to achieving an overall carbon-neutral cycle. In a carbon-constrained global economy, our forests and wood products are the ultimate renewable resource.

Australian forestry. Planning for tomorrow, today.

Did you know?

- Australia's forest industries use drones to obtain improved forest data. They use this data to improve how forests are managed.
- Drones are also used for improved fire control and helping manage threatened species in forests.
- Drones are even being tested to see if they can be used to plant trees.

Source: <https://www.agriculture.gov.au/agriculture-land/forestry/planning-tomorrow/innovation-and-technology>